

Table 4. Ablation analysis on BRCA.

Variant	ACC	NMI	ARI	PUR
STCKLA	89.42	73.30	72.33	89.42
w/o module-guided input selection	88.82	70.77	71.93	88.82
w/o knowledge alignment	81.67	66.44	58.86	81.67
w/o residual feature adapter	78.54	51.48	53.33	78.54
w/o expression reconstruction	87.93	70.65	70.54	87.93
w/o supervised classification	88.08	71.04	70.11	88.08
w/o prototype learning	59.76	51.81	34.96	79.43
w/o cross-view consistency	79.58	57.91	53.22	81.22